Challenge 5: Erdős–Hajnal Conjecture

Definition 1 (Cliques and Independent sets)**.** Given a simple graph G, a set of vertices of G is a *clique* if its vertices are pairwise adjacent. It is an *independent set* if its vertices are pairwise non-adjacent.

Definition 2 (H-free graphs)**.** Given graphs G and H, we say that G is *H-free* if it does not contain an induced subgraph isomorphic to H.

Conjecture 1.1 (Erdős–Hajnal Conjecture [1])**.** For each graph H, there is a constant $c_H > 0$ such that every H-free graph G with n vertices has a clique or independent set of size at least n^{c_H}.

Given $r \in \mathbb{N}$, we denote the path with r vertices by P_r. We are interested in the following parametrized version of the conjecture, where $H = P_r$ is a path.

Conjecture 1.2 (Erdős–Hajnal Conjecture for paths)**.** Let $r \in \mathbb{N}$. There is some $c_r > 0$ such that every P_r-free graph G with n vertices has a clique or independent set of size n^{c_r}.

The conjecture has been proved for $r = 5$ by Nguyen, Tung, Scott and Seymour in [2]

References

[1] P. Erdös and A. Hajnal. "Ramsey-type theorems". In: *Discrete Applied Mathematics* 25.1 (1989), pp. 37–52. ISSN: 0166-218X. DOI: `https://doi.org/10.1016/0166-218X(89)90045-0`. URL: `https://www.sciencedirect.com/science/article/pii/0166218X89900450`.

[2] T. Nguyen, A. Scott, and P. Seymour. "Induced subgraph density. VII. The five-vertex path". In: *Proceedings of the London Mathematical Society* 132.3 (2026), e70133.